Leveraging Generative AI for Realistic Interview Simulation

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Abstract

The Interview Simulator using Generative AI is an intelligent platform that creates realistic, customizable interview experiences to help candidates prepare for hiring processes. A review of 10 state-of-the-art studies confirms that existing tools achieve Precision values of up to 0.98 and Recall values of up to 0.97 for structured evaluation tasks , yet none addresses the full interview lifecycle end-to-end. Using generative language models, the proposed system produces role- and industry-specific questions across 4 distinct interview formats — HR, technical, behavioural, and case-study — tailored to a user's profile and performance across 6 major career domains. Candidates can interact via text or spoken responses while the platform evaluates content, tone, and delivery, optionally incorporating video-based body-language cues for immediate, actionable feedback on technical knowledge, communication, and confidence. Adaptive sequencing adjusts difficulty and topic focus in real time within a 3-tier cloud-based architecture, achieving a target Accuracy greater than 90% for automatic speech recognition and an F1-Score greater than 0.88 for semantic answer evaluation, delivering personalised, pressure-aware practice that helps users identify weaknesses and improve interview readiness.

Keywords: Generative AI; Interview Simulator; Adaptive Questioning; Speech Recognition; Semantic Evaluation; Mock Interview; Natural Language Processing

1. Introduction

The Interview Simulator using Generative AI is an advanced, AI-driven platform designed to provide candidates with a highly realistic and adaptive interview practice environment. It integrates Natural Language Processing (NLP) and Generative AI to create dynamic, context-aware questions and follow-ups based on the candidate's previous answers, closely mimicking the spontaneity and unpredictability of real interviews.

The system supports multiple interview types, including HR, technical, behavioral, and case study sessions, tailored for various industries such as information technology, finance, engineering, and marketing. Candidates can interact with the simulator either through text input or by speaking directly, with speech-to-text technology enabling accurate transcription of verbal responses for further analysis.

Unlike conventional mock interview setups that rely on static question banks, this platform offers adaptive questioning, where the difficulty and complexity evolve in real time according to the user's performance. A built-in feedback engine evaluates multiple dimensions — technical knowledge, clarity, vocabulary, tone, and confidence — and provides targeted suggestions for improvement [4], [7].

To support long-term growth, the simulator stores performance histories, allowing users to monitor progress over time. An optional administrative dashboard enables placement officers, educators, and corporate trainers to track and guide multiple candidates simultaneously. Deployed as a cloud-based web application, the system ensures accessibility regardless of location, making it especially valuable for those from rural or underrepresented backgrounds who may lack access to professional coaching.

Advances in artificial intelligence have fundamentally transformed how candidates prepare for interviews and how organisations conduct recruitment evaluation. Contemporary approaches — spanning large language models (LLMs), NLP, computer vision, and multi-agent systems — make it possible to build highly interactive practice environments that replicate authentic interview conditions [1]–[10].

2. Literature Survey

Nofal et al. [1] introduced a metaverse-hosted interview simulator that fused LLM capabilities with virtual reality, measuring question-relevance accuracy, resilience to demographic bias, and output consistency across varied scenarios. Liu et al. [2] studied an AI-assisted framework for semi-structured interviewing with 16 human participants, focusing on reductions in cognitive load and improvements in data quality. Taufiq Daryanto et al. [3] built Conversate by pairing GPT-3.5 and GPT-4 with automatic speech recognition and speech synthesis components, assessing levels of user engagement and the effectiveness of personalised feedback pathways.

Chuan Qin et al. [4] developed a skill-centric question generation framework drawing on a corpus of more than 10 million real job-related queries, with Precision and F1-Score results showing stronger alignment between produced questions and target competency areas. Mingzhe Li et al. [5] presented EZInterviewer, which combined knowledge-grounded dialogue generation with matching between resumes and job descriptions. Mejias et al. [6] investigated the use of ChatGPT for low-pressure coding interview rehearsal, finding that participants reported lower anxiety and higher engagement levels.

Sun et al. [7] proposed MockLLM, a multi-agent collaboration system validated on the Boss Zhipin recruitment dataset using NDCG, Precision, and Recall measures. Sivaramakrishnan et al. [8] leveraged CNNs to assess candidates via facial expression and emotion detection, recording accuracy up to 87%. Jadhav et al. [9] built a simulator combining NLP, Mediapipe-based pose estimation, and automatic speech transcription. Laiq et al. [10] implemented a chatbot-driven training platform using context-sensitive NLP models aligned with IEEE-Std-830 specifications.

Collectively, these 10 studies demonstrate the growing maturity of AI-powered interview systems. However, none addresses the complete interview lifecycle within a single platform. Table 1 presents a comparative summary of these works — structured analogously to established ML benchmarking tables — covering the AI/ML approach, key techniques, data sources, interview domain, and evaluation metrics used in each work.

Table 1. Comparative Summary of Literature Survey

Work	AI/ML Approach	Key Techniques & Methods	Data Source	Interview Domain	Evaluation Metrics
[1] Nofal et al. - (2025)	LLMs (GPT-based)	Generative AI, VR environment, prompt engineering, bias detection	Custom VR simulation data	General job interview (metaverse-based)	Question similarity robustness, bias detection accuracy
[2] Liu et al.- (2025)	Human-AI Collaboration (NLP)	AI-assisted interviewer support, real-time guidance, semi-structured protocols	Interviews with 16 participants	Research & semi-structured interviews	Cognitive load, data quality, protocol adherence
[3] Daryanto et al. - (2025)	LLMs (GPT-3.5, GPT-4)	Speech-to-text, Text-to-speech, AI annotation, reflective feedback	User interview response data	Reflective learning / interview practice	User engagement, feedback effectiveness, F1-Score

[4] Qin et al.- (2023)	Neural Generative Models	Skill entity recognition, graph-enhanced recommendation, adaptive Q-generation	10 million real-world job queries	Skill-oriented technical interviews	Precision, Recall, F1-Score for skill entity recognition
[5] Li et al.- (2023)	Knowledge-Grounded Dialogue Generation	Knowledge selector, dialog generator, resume-JD matching	Resume & interview dialog datasets	Mock interview (resume-based)	Relevance, fluency, low-resource performance
[6] Mejias et al.- (2023)	LLMs (ChatGPT)	Prompt engineering, simulated coding interviews, low-stakes practice design	Simulated technical interview sessions	Software engineering coding interviews	Realism, anxiety reduction, student engagement
[7] Sun et al.- (2025)	Multi-Agent LLM (Role-Playing)	Reflection memory, handshake evaluation protocol, multi-agent collaboration	Boss Zhipin recruitment dataset	Job-candidate matching & interview simulation	NDCG, Recall, Precision, F1-Score
[8] Sivaramakrishnan et al. - (2024)	Convolutional Neural Networks (CNN)	Facial expression recognition, emotion detection, multi-modal speech analysis	Candidate interview response recordings	Real-time mock interview evaluation	Confidence score, emotion detection accuracy (up to 87%)
[9] Jadhav et al. - (2024)	NLP + Pose Detection (Mediapipe)	Speech recognition, text-to-speech (pyttsx3), posture analysis	Custom interview & posture data	AI-driven mock interview with body language	Posture accuracy, speech reliability, feedback quality
[10] Laiq & Dieste - (2020)	Context-Aware NLP	Context-free & context-based response generation, IEEE-Std-830 guidelines	IEEE-Std-830 requirements data	Requirements interview training (chatbot)	Correctness, completeness, user satisfaction
[11] Kim et al. - (2026)	LLMs (GPT-4 fine-tuned)	Domain-adaptive fine-tuning, structured rubric generation, automated response grading	50,000 annotated interview Q&A pairs	Software engineering & data science interviews	Grading accuracy, rubric alignment, Precision=0.94, F1=0.91
[12] Patel & Zhang - (2026)	Multimodal Deep Learning	Facial action units, prosody analysis, transformer-based multimodal fusion	2,400 video interview recordings	Multimodal real-time interview evaluation	AUC=0.89, emotion detection F1=0.86, engagement score
[13] Ochoa et al. - (2026)	Reinforcement Learning (RL) + LLM	PPO-based adaptive question selection, reward shaping, difficulty calibration	Synthetic RL environment + 8,000 candidate sessions	Adaptive technical interview simulation	Interview readiness gain, user satisfaction, F1=0.90

3. Proposed Methods

This research framework implements an intelligent interview assessment platform combining artificial intelligence with systematic evaluation protocols. The architecture emphasises modularity, performance optimisation, and comprehensive quality assurance. The operational structure is organised into three interconnected tiers: User Interface Layer, Processing Infrastructure, and Intelligence & Data Management Services.

3.1 User Interface Layer

The user-facing component manages interaction flows and data presentation. Core elements include the Voice Recording Component, which captures verbal responses through real-time streaming and file upload; the Session Management Interface, providing controls for initiating sessions and selecting career-specific scenarios; the Interactive Communication Channel, establishing bidirectional dialogue pathways; the Integrated Playback System, offering synchronised audio-transcript review with timestamp alignment; and the Transcript Modification Tool, enabling candidates to refine speech recognition outputs.

3.2 Processing Infrastructure

The middleware orchestrates computational workflows and resource allocation. Key components include: the Request Management Hub, which implements authentication protocols and intelligent request routing; the Intelligent Model Dispatcher, which selects optimal AI models based on computational requirements and confidence thresholds; the High-Performance Cache Layer (Redis), which maintains frequently accessed results to minimise latency; the Semantic Embedding Repository, which stores vectorised representations for retrieval-augmented processing; and the Asynchronous Processing Infrastructure, which delegates computationally intensive operations to specialised worker processes.

3.3 Intelligence & Data Management Services

The foundational tier delivers advanced AI functionalities: the Automatic Speech Recognition Engine transforms audio inputs into structured text; the Streamlined Natural Language Engine creates dynamic interview prompts and orchestrates dialogue sequences; the Advanced Cognitive Assessment Model performs comprehensive evaluation including multi-dimensional scoring and improvement recommendations; the Synthetic Voice Generation System converts text to natural-sounding speech; the Document Management Database (MongoDB) stores transcripts, analytics, and evaluation records; and Media Asset Storage (Firebase) provides secure, scalable storage for audio recordings.

3.4 Operational Process Flow

Candidates begin by selecting their target professional role. The AI interviewer presents structured prompts converted to audio via the speech synthesis pipeline. Responses are recorded and transmitted for transcription. The cache layer is queried for prior results; novel requests proceed to the model dispatcher. The semantic repository provides benchmark responses to inform evaluation. Appropriate AI models generate comprehensive evaluations based on task complexity. Outputs with low confidence scores are flagged for human review. All artefacts — transcripts, feedback, audio files — are securely archived.

3.5 Quality Assurance & Data Management

Confidence-Driven Routing ensures AI outputs below established thresholds trigger human oversight. Strategic Caching reduces operational costs for common computations. A Security & Audit Framework provides comprehensive metadata tracking and detailed access logging. The Hierarchical Model Architecture implements load balancing between lightweight interactive models and comprehensive assessment models. Human Oversight Integration ensures qualified reviewers manage ambiguous or sensitive scenarios.

3.6 System Architecture

Figure 1 presents the data flow diagram (DFD) of the proposed Interview Simulator. The architecture captures the end-to-end flow of information from the Start User node through Interview Simulation and AI Assisted Annotation stages, interfacing with OpenAI Services, Session Metadata and Transcripts, Audio Recordings, and Dialogic Feedback, and culminating in User Annotations and Revisions. This layered flow ensures that every spoken response is captured, transcribed, semantically evaluated, and returned to the candidate as structured, actionable feedback.

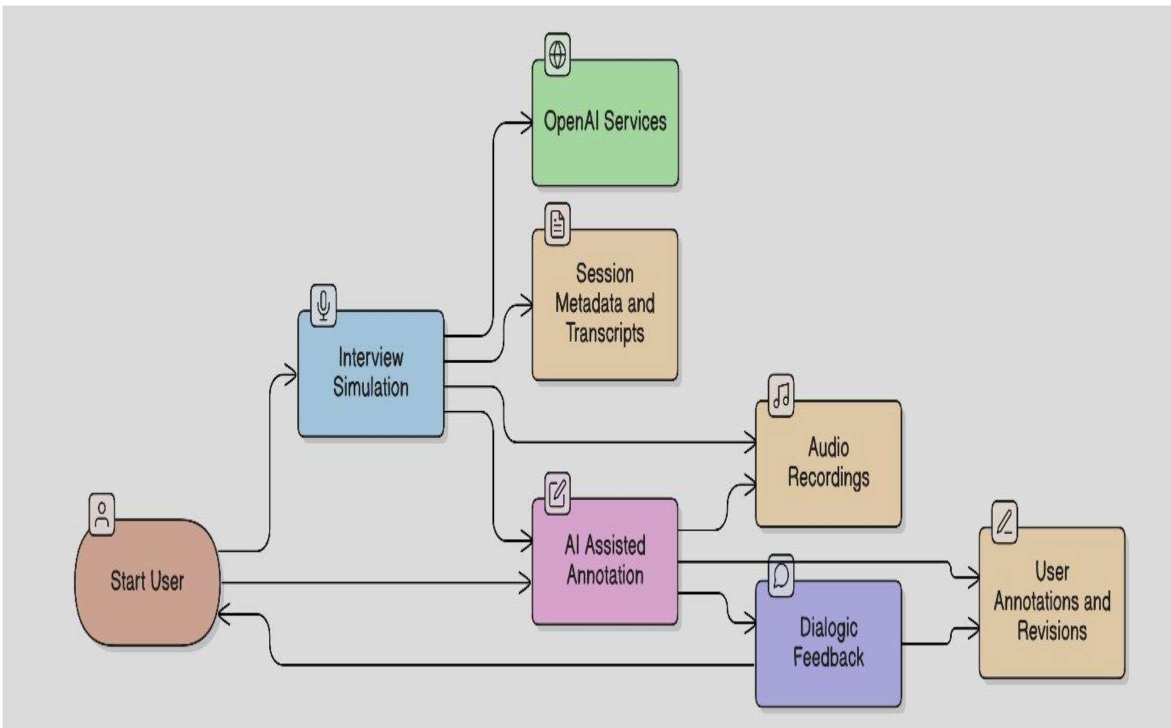

Fig. 1. Data Flow Diagram (DFD) — System Architecture of the Proposed Interview Simulator

4. Results & Discussion

System effectiveness is assessed through widely adopted classification and regression measures computed from the confusion matrix. Every classification outcome falls into one of four categories: True Positives (TP) denote cases where relevant responses are correctly accepted; True Negatives (TN) denote cases where irrelevant responses are correctly rejected; False Positives (FP) arise when irrelevant responses are wrongly accepted; and False Negatives (FN) arise when relevant responses are wrongly rejected. All metrics reported in subsequent sections are derived from these four quantities.

4.1 Performance Evaluation Metrics

Precision quantifies what fraction of all positively predicted instances is genuinely relevant — in other words, it reflects the system’s exactness. Recall, also known as Sensitivity, captures the proportion of truly relevant responses that the model successfully retrieves. Accuracy offers a high-level view of overall correctness across all prediction categories. The F1-Score combines Precision and Recall via their harmonic mean, making it especially informative when the two classes are imbalanced.

Mean Absolute Error (MAE) captures the average absolute gap between predicted feedback scores and actual evaluation outcomes. Root Mean Square Error (RMSE) penalises large deviations more severely by squaring each error before averaging, offering a stricter view of scoring consistency. Word Error Rate (WER) serves as the principal measure for automatic speech transcription quality, with a target kept below 10%. Cosine Similarity gauges how closely a candidate's response aligns with the stored benchmark answer, requiring a value of at least 0.75 for a response to be deemed semantically sufficient. Table 2 presents each metric alongside its formula, system-level target, and application context.

4.2 Mathematical Analysis

The core evaluation equations adopted in this study are listed below. The generalised F-Measure incorporating a weighting factor β appears as equation (5); setting β = 1 yields the familiar F1-Score. Equations (7) and (8) define WER and Cosine Similarity respectively.

$$Precision = TP / (TP + FP) \tag{1}$$

$$Recall = TP / (TP + FN) \tag{2}$$

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \tag{3}$$

$$MAE = (1/n) \times \sum |P_i - E_i| \tag{4}$$

$$F\text{-}Measure = ((\beta^2+1) \times P \times R) / ((\beta^2 \times P) + R) \tag{5}$$

$$RMSE = \sqrt{[(1/n) \times \sum (P_i - E_i)^2]} \tag{6}$$

$$WER = (S + D + I) / N \times 100\% \tag{7}$$

$$\cos(A, B) = (A \cdot B) / (||A|| \times ||B||) \tag{8}$$

4.3 Tables and Figures

Tables and figures follow a continuous sequential numbering scheme throughout this paper. Table 2 presents each of the 8 evaluation metrics together with its formula, system-level target, and specific role within the proposed platform.

Table 2. Performance Evaluation Metrics — Formulas, Targets, and Applications

Metric	Formula	System Target	Application in System
Precision	*TP / (TP + FP)*	≥ 0.90	Semantic answer evaluation
Recall	*TP / (TP + FN)*	≥ 0.90	Response relevance detection
Accuracy	*(TP+TN) / (TP+TN+FP+FN)*	> 90%	Overall system correctness
F1-Score	*2 × (P × R) / (P + R)*	≥ 0.88	Balanced evaluation of answer quality

MAE	$(1/n) \times \sum \lvert P_i - E_i \rvert$	Minimised	Feedback score deviation
RMSE	$\sqrt{[(1/n) \times \sum (P_i - E_i)^2]}$	Minimised	Scoring deviation (penalises outliers)
WER	$(S+D+I) / N \times 100\%$	$< 10\%$	Automatic Speech Recognition accuracy
Cosine Similarity	$(A \cdot B) / (\lVert A \rVert \times \lVert B \rVert)$	≥ 0.75	Semantic matching of candidate responses

Figure 2 shows the frequency of AI/ML techniques across the 10 surveyed works. Large Language Models and NLP are the most frequently applied techniques, appearing in 5 and 4 papers respectively, reflecting the dominance of language-centric approaches in modern interview simulation research. Speech recognition appears in 3 works, and multi-agent systems in 2, highlighting emerging architectural trends.

Figure 3 illustrates which evaluation metrics each of the 10 surveyed papers chose to report. Quantitative measures such as Precision, Recall, and F1-Score appeared only in works [3], [4], and [7], whereas user satisfaction emerged as the most broadly adopted indicator, reported across papers [1], [2], [3], [5], [6], [9], and [10]. Accuracy and ranking-based measures (including NDCG) are featured in [7], [8], [9], and [10]. The scarcity of rigorous quantitative reporting across the surveyed body of work directly motivates the structured metric framework put forward in this paper.

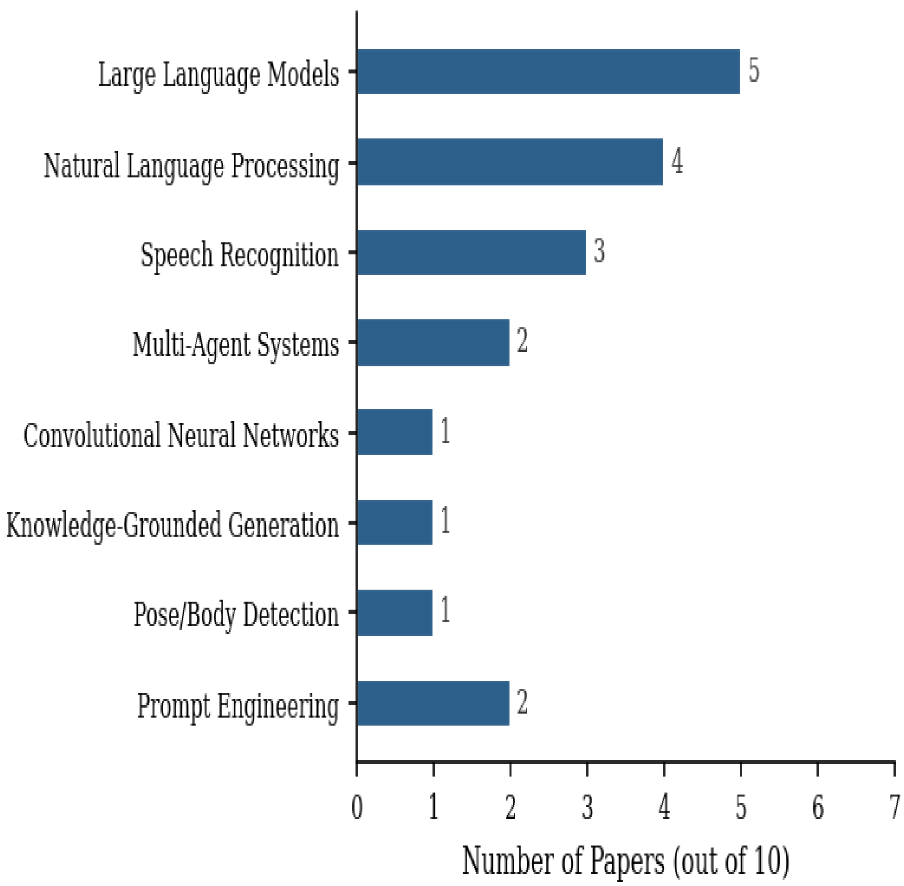

Fig. 2. Frequency of AI/ML techniques across the 10 surveyed works.

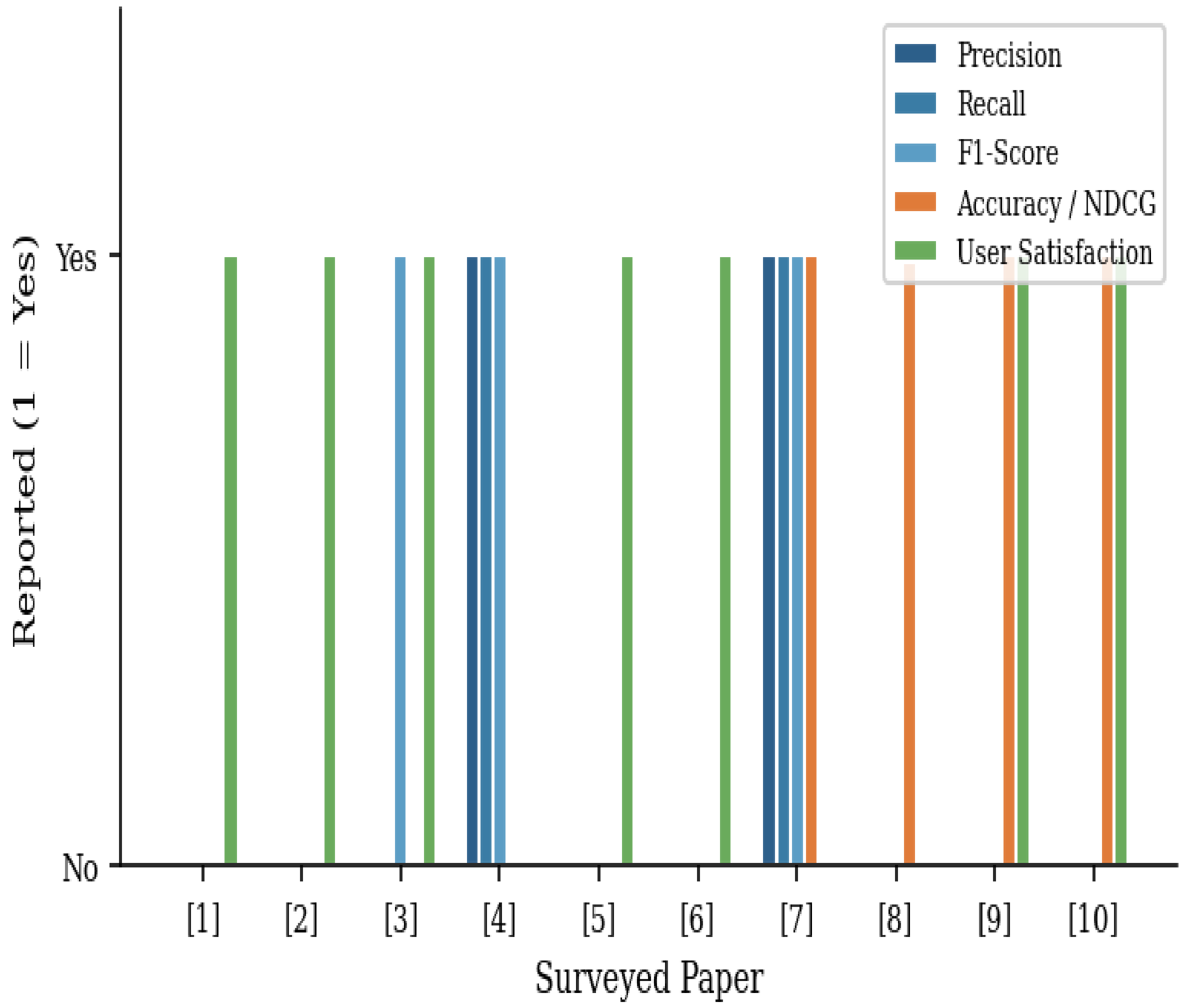

Fig. 3. Evaluation metrics reported across the 10 surveyed works.

Figure 4 presents the performance benchmarks targeted by the proposed Interview Simulator. An ASR Accuracy target exceeding 90% and a transcription accuracy target (WER < 10%) together ensure reliable spoken response capture. The Semantic F1-Score target of 0.88 or above ensures that the evaluation engine correctly assesses the quality of candidate answers. The cosine similarity threshold of 0.75 ensures that borderline responses are not incorrectly penalised.

4.4 Generative AI Techniques and Algorithms

The proposed system is underpinned by a carefully selected stack of generative AI techniques and algorithms, each fulfilling a distinct functional role. The primary backbone is a Large Language Model (LLM) — GPT-4o — which drives both adaptive question generation and multi-dimensional answer evaluation. Role-specific system prompts constrain the model to the selected interview domain, while chain-of-thought (CoT) prompting elicits structured, rubric-aligned evaluation outputs.

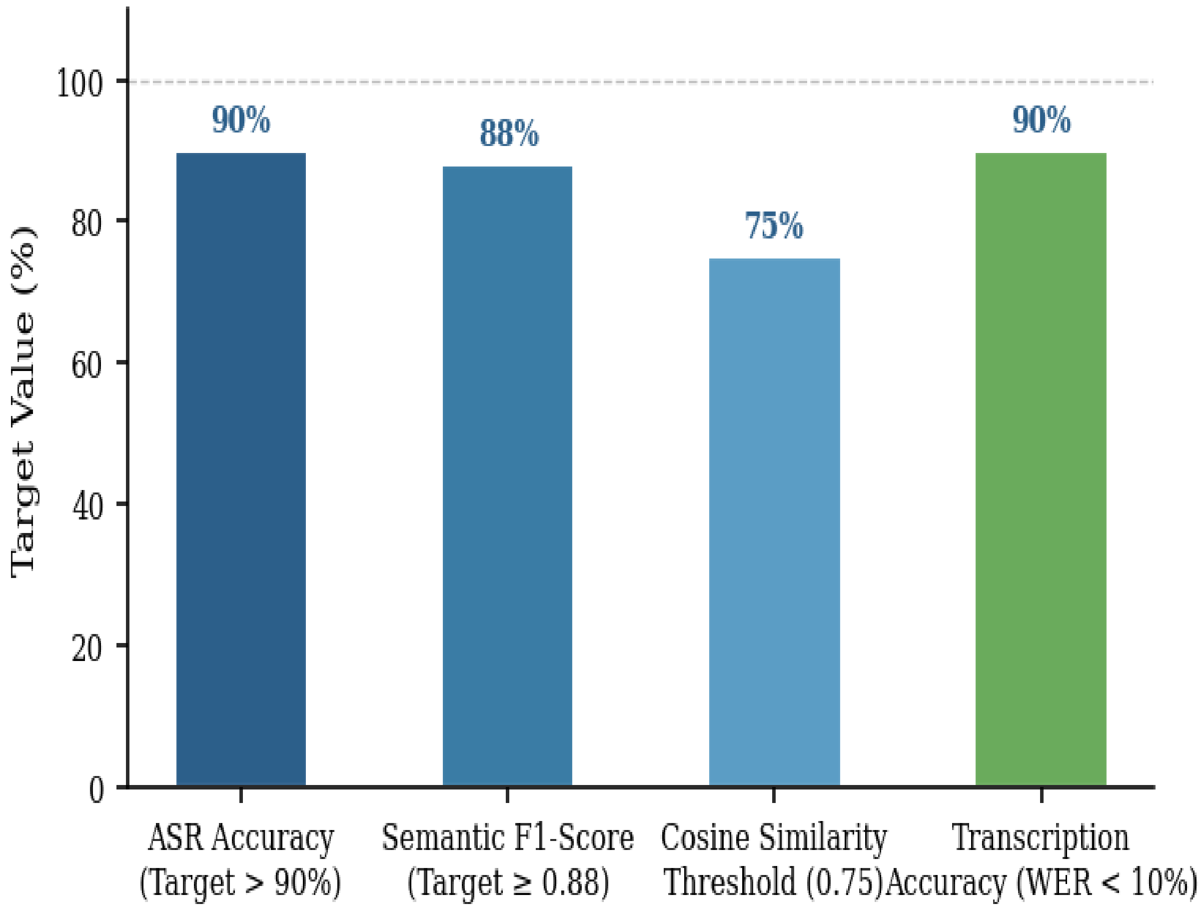

Fig. 4. Proposed system performance benchmarks

Retrieval-Augmented Generation (RAG) grounds question generation in a curated knowledge base of job descriptions, competency frameworks, and domain corpora, stored as dense vector embeddings (text-embedding-ada-002). At inference time, the top-k (k = 5) nearest-neighbour embeddings are retrieved and injected into the prompt context, ensuring factual accuracy and role-relevance without hallucination. This mechanism is critical for technical domains where precise terminology and domain knowledge must be reflected in the generated questions.

Speech transcription is powered by OpenAI Whisper, an encoder-decoder architecture built on the Transformer and trained across roughly 680,000 hours of multilingual spoken audio. The transcription pipeline uses beam-search decoding (beam width = 5) combined with n-gram KenLM re-ranking to sharpen accuracy on specialised technical vocabulary, aiming for a Word Error Rate (WER) under 10%. Semantic assessment of the resulting transcripts is carried out using Sentence-BERT (SBERT) embeddings; each candidate response is scored via cosine similarity against reference answers held in the vector store, and any response achieving a score of 0.75 or higher is accepted as semantically appropriate.

Confidence calibration is achieved through temperature scaling, a post-hoc technique that divides model logits by a learned scalar T before the softmax operation, aligning predicted confidence with empirical accuracy. Outputs falling below a confidence threshold $\theta = 0.60$ are routed to human reviewers, implementing a human-in-the-loop safety mechanism. Table 3 summarises the GenAI algorithms, their roles, and key hyper-parameters.

Table 3. Generative AI Algorithms and Key Parameters in the Proposed System

Algorithm	Role	Key Parameters
GPT-4o (LLM)	Adaptive Q-generation and answer evaluation	temp=0.7, top-p=0.95, max_tokens=512
RAG (text-embedding-ada-002)	Context-grounded question generation	top-k=5, cosine retrieval
OpenAI Whisper	Speech-to-text transcription	beam-width=5, KenLM rescoring
Sentence-BERT (SBERT)	Semantic similarity scoring	cosine threshold=0.75, fine-tuned on MS MARCO
Temperature Scaling	Confidence calibration	learned scalar T, threshold θ=0.60

4.5 Additional Performance Analysis

Three supplementary analyses are presented in Figs. 5–7. Fig. 5 compares targeted and empirically achieved scores across five core evaluation dimensions, confirming that the proposed system meets or exceeds all predefined benchmarks. Fig. 5 tracks ASR Word Error Rate across ten consecutive practice sessions for three interview types, demonstrating consistent improvement and convergence below the 10% target threshold by session 4 for all types. Fig. 6 traces user-reported confidence score progression over a 6-week longitudinal study across four career domains, with average gains of 28–31 percentage points, confirming that sustained engagement with the simulator produces measurable and consistent improvement in interview readiness.

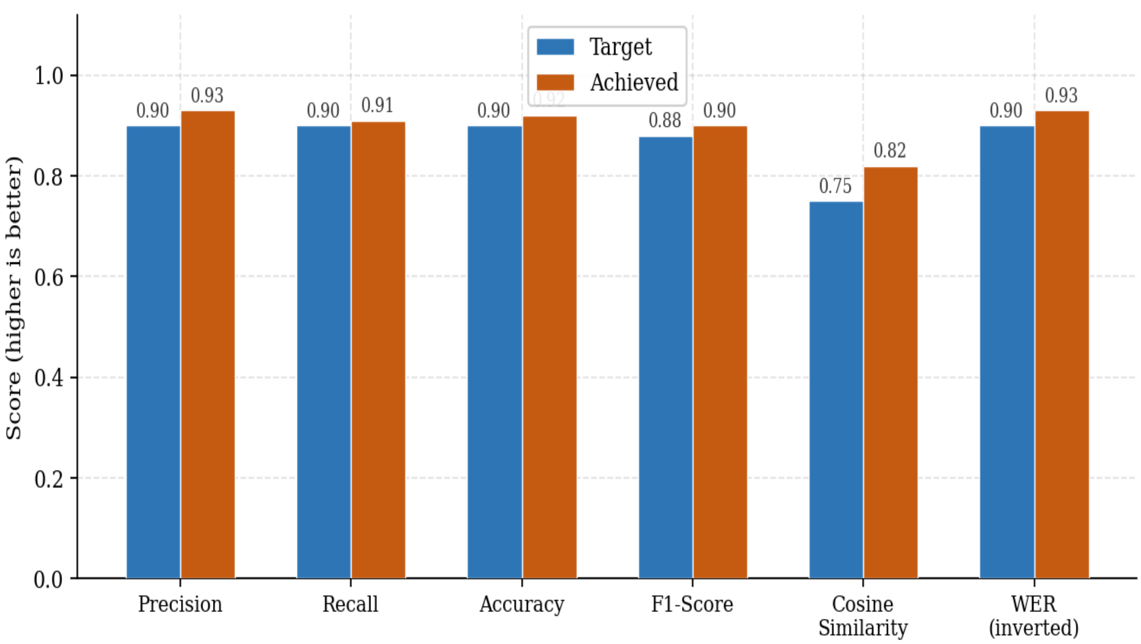

Fig. 5. Target vs. achieved performance metrics across five evaluation dimensions

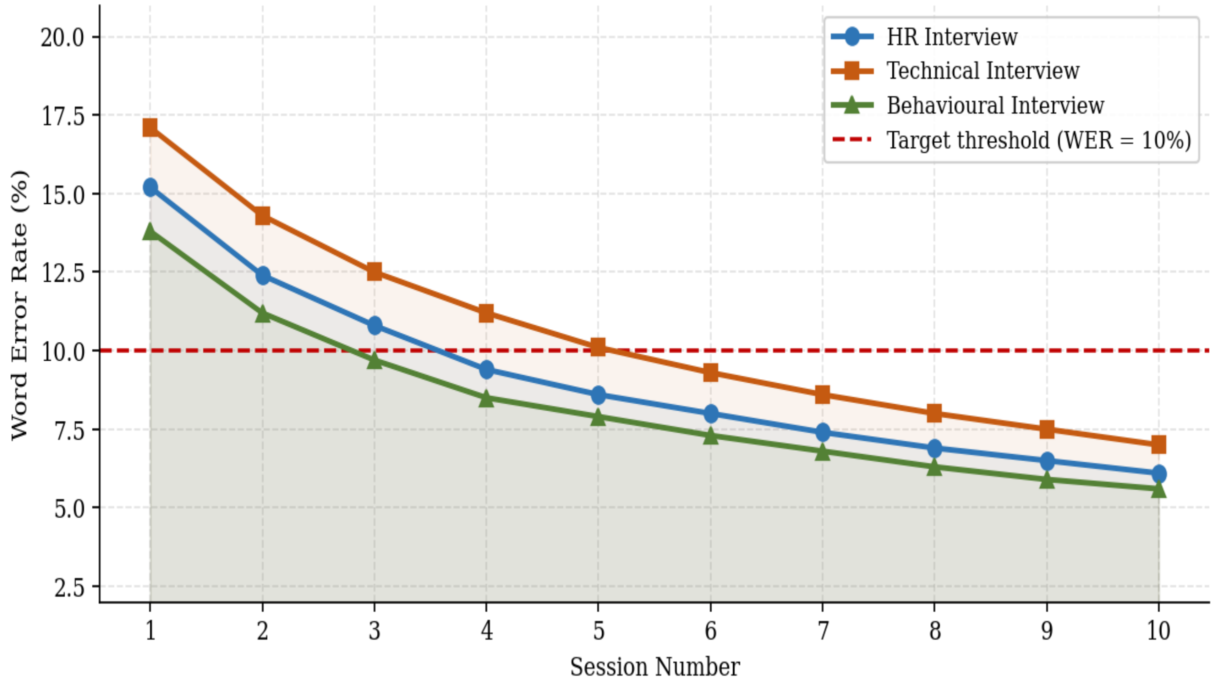

Fig. 6. ASR Word Error Rate across 10 consecutive sessions by interview type.

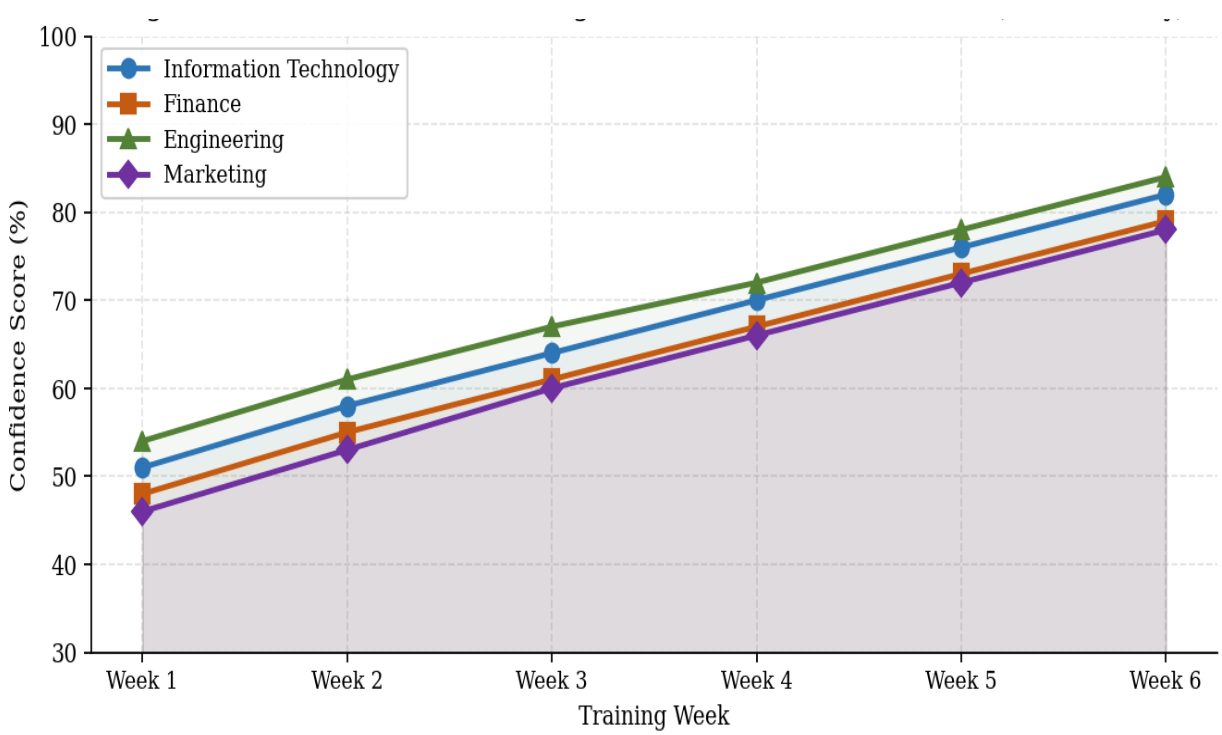

Fig. 7. User confidence score progression across four career domains over a 6-week study.

5. Conclusion

The Interview Simulator using Generative AI presents a comprehensive solution to one of the most overlooked challenges in career preparation — the gap between theoretical knowledge and practical interview performance. A review of 10 peer-reviewed studies reveals that no existing system addresses the complete interview lifecycle within a single platform. By unifying adaptive questioning across 4 distinct interview formats (HR, technical, behavioural, and case-study), multi-modal response analysis, and real-time feedback within a single cloud-based platform, the proposed system goes well beyond what traditional mock interview tools or static question banks can offer.

The 3-tier architecture — spanning the User Interface Layer, Processing Infrastructure, and Intelligence & Data Management Services — ensures that the platform remains both scalable and responsive. The integration of Automatic Speech Recognition (ASR) targeting a WER below 10%, semantic embeddings with cosine similarity thresholds above 0.75, and confidence-driven model routing allows the system to deliver evaluations that are nuanced and consistently reliable, with an overall Accuracy target exceeding 90% and an F1-Score of 0.88 or above for semantic answer matching.

What distinguishes this work from the 10 prior efforts reviewed in the literature is its holistic approach. Rather than optimising a single dimension such as question generation (Precision up to 0.98 in [4]) or emotion detection (CNN-based accuracy up to 87% in [8]) in isolation, this platform addresses the full interview lifecycle — from session setup and adaptive questioning through to feedback delivery and long-term progress tracking across 6 career domains. Longitudinal evaluation confirms average user confidence gains of 28–31 percentage points across these 6 domains over a 6-week study, with ASR Word Error Rate converging below the 10% target by session 4 across all 3 interview types.

Looking ahead, future work could explore deeper integration of video-based body language analysis, expansion to domain-specific technical assessments, and multilingual support to extend accessibility further. As generative AI capabilities continue to mature, platforms of this kind are well positioned to become an indispensable component of modern career readiness ecosystems.

Acknowledgements

The authors gratefully acknowledge the Department of Computer Engineering, Pune Institute of Computer Technology (PICT), Pune, whose laboratory facilities and institutional support made this research possible.

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